Artificial Intelligence Technology and Application

Machine Learning Lab Guide

Student Version

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1 E-commerce Website User Group Analysis

1.1 Introduction

1.1.1 About This Lab

This experiment performs modeling based on the *k*-means algorithm by using the virtual dataset automatically generated by **sklearn** to obtain user categories. It is a clustering experiment, which can find out the method for selecting the optimal *k* value and observe the effect in a visualized manner.

1.2 Experiment Code

1.2.1 Using sklearn for Modeling

Step 1 Import libraries.

```
import numpy as np
import matplotlib.pyplot as plt
```

Step 2 Create a dataset.

Create virtual data for the algorithm model.

```
from sklearn.datasets import make_blobs
```

The built-in tool of **sklearn** is used to create the virtual data, which is scientific and conforms to a normal distribution. Parameter settings are as follows:

- **n_samples**: set to **2000**, indicating that 2000 sample points are set.
- **centers**: set to **2**, indicating that the data actually has two centers.
- **n_features**: set to **2**, indicating the number of features.

For ease of illustration in the coordinate system, only two features are used.

Step 3 Apply the *k*-means algorithm.

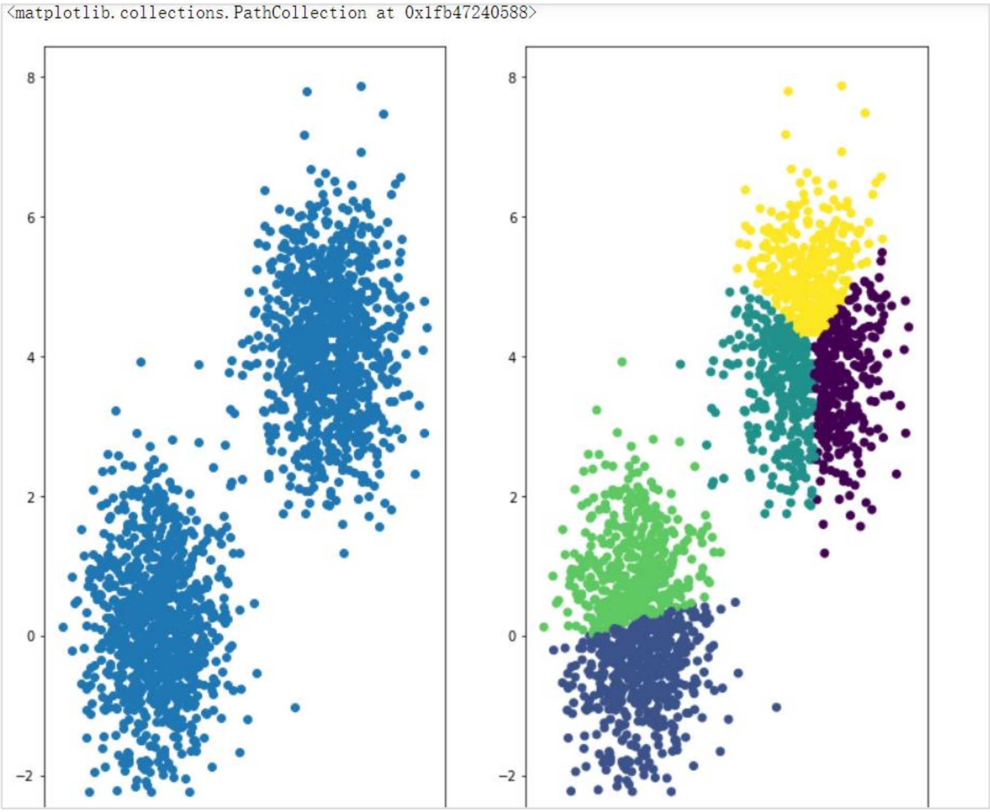
```
from sklearn.cluster import KMeans
```

n_clusters=5: indicates that five data clusters are expected. However, there are only two data categories.

Step 4 Visualize the output.

```
import matplotlib.pyplot as plt
```

Output:

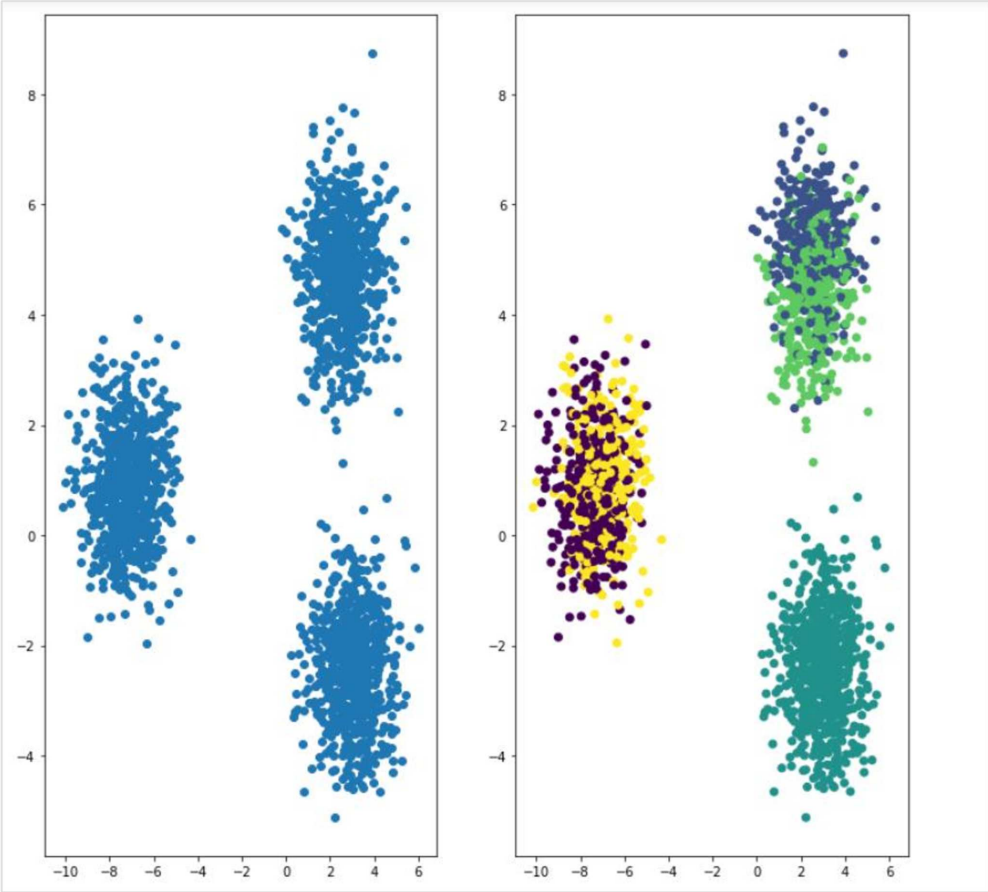


Different data is generated each time. Therefore, the output diagram may be different from that in the lab. To generate the same data, add the **random_state** parameter during data generation.

In this example, **random_state** is set to **3**. In this way, the same data can be generated for the same data input.

Step 5 Use more features for comparison.

In this example, ten features are used to generate data, **random_state** is set to **30**, and there are three categories in theory.



----End

1.2.2 Selecting the Optimal k Value

In the preceding steps, the *k* value is manually set. In actual environments, the number of centers is unknown. Therefore, you need to find the optimal *k* value.

```
import random
```

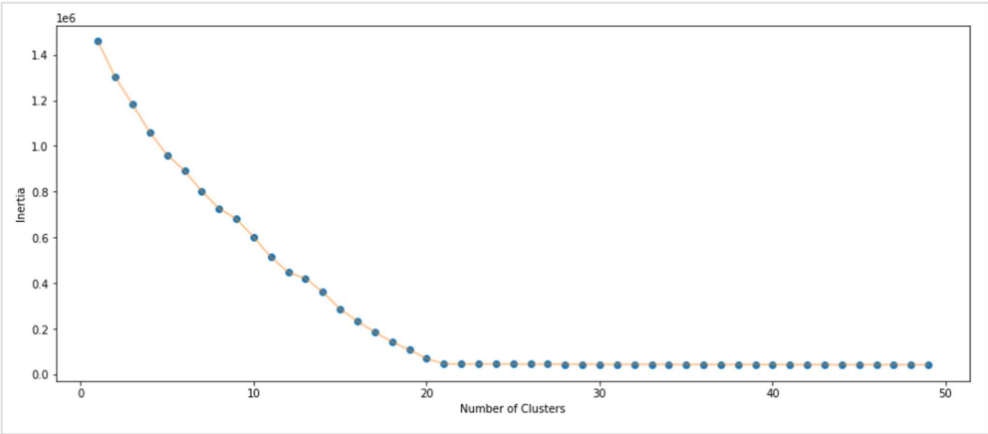
First, generate two random numbers ranging from 1 to 30 (indicating that the number of true centers in the data is unknown), and use a random number of features.

```

```

Then, perform *k*-means clustering by using a recursive method. The **.inertia_** attribute returns the distance from the attribute point to the center.

```

```

Visualize the result by using a visualization tool.



The result varies each time due to impact of the random numbers. As shown in the preceding figure, the turning point appears at the position corresponding to the value 21. Therefore, 21 is the optimal *k* value.